

Congruence & Similarity

Quick reference sheet · GCSE Maths, Geometry & Measures · section 5 is Higher tier

1. Congruent vs similar

Congruent = exactly the same shape **and** size (may be turned or flipped). All matching sides and angles are equal.

Similar = same shape, different size (one is an enlargement of the other). Matching angles are equal; matching sides are all in the same ratio.

2. Conditions for congruent triangles

Condition	What must match
SSS	All three pairs of sides equal
SAS	Two sides and the angle <i>between</i> them (included angle)
ASA / AAS	Two angles and a corresponding side
RHS	Right angle, equal hypotenuses, and one other equal side

AAA does **not** prove congruence (only similarity). SSA does **not** prove congruence either. In a proof: state three facts, give a reason for each, then name the condition.

3. Similar shapes — missing lengths

1) Find a pair of matching sides where both lengths are known. 2) Scale factor = new length $\div$ original length. 3) Multiply by the SF to go from small to large; divide to go from large to small.

To test whether two shapes are similar, check that the ratio is the **same for every pair** of matching sides.

4. Similar triangles inside each other

If DE is parallel to BC (D on AB, E on AC), triangles ADE and ABC are similar: angle A is shared and the parallel lines make corresponding angles equal. Use the **whole** side $AB = AD + DB$ for the large triangle. Sketch the two triangles separately to match up sides.

5. Area & volume scale factors (Higher)

Length SF	Area SF (incl. surface area)	Volume SF
k	k^2	k^3
2	4	8
3	9	27

Given an area ratio? **Square-root** it to get the length SF. Given a volume ratio? **Cube-root** it. Always get back to the length SF first, then square or cube as needed.

6. Common mistakes

Using DB instead of the whole side AB in nested triangles.

Multiplying an area or volume by the length scale factor instead of k^2 or k^3 .

Pairing up the wrong sides — match sides using equal angles, not position on the page.

Writing a congruence proof with no reasons, or using AAA as a condition.

Dividing the wrong way round: $SF = \text{new} \div \text{original}$, and check the answer is bigger or smaller as expected.